Goals:

- Definition of interior points and open sets in a metric space;
- Examples of open sets in different metric spaces;
- Main properties of open sets (The empty set and the whole space are open, a finite intersection of open sets is open, an arbitrary intersection of open sets is open);
- A nonfinite intersection of open sets may not be open.

Review:

- Definition of metric space;
- The following metrics: the Euclidean metric, the d_∞ metric on $\mathbb{R}^2$, the discrete metric on a set;
- Unions and intersections of families of sets.

Definition. *Let (X, d) be a metric space and $A \subseteq X$.*

- *A point $x_0 \in X$ is an* **interior point** *of A if there exists $r > 0$ such that $B_r(x_0) \subseteq A$.*
- *The* **interior** *of A is defined as the set of all interior points of A, that is*

$$\mathring{A} = int(A) = \{x_0 \in X \mid x_0 \text{ is an interior point of } A\}.$$

- *We say that A is* **open** *if every point of A is an interior point, that is*

$$\forall x \in A,\ \exists r > 0 \ s.t\ B_r(x) \subseteq A.$$

Example 1 The set $A =]0, 1[$ is open in $(\mathbb{R}, d)$ where d is the Euclidean metric

Example 2 The set $A = [0, 1[$ is not open in $(\mathbb{R}, d)$ where d is the Euclidean metric

Example 3 The set $A =]0, \infty[\times]0, \infty[= \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 > 0, x_2 > 0\}$ is open in $(\mathbb{R}^2, d_\infty)$

Exercise 1 (Any set is open in the discrete metric!) Let $X \neq \varnothing$ be equipped with the discrete metric and $A \subseteq X$. Then, A is open.

Proposition (In a metric space open balls are open indeed!)**.** *Let* (X, d) *be a metric space,* $x_0 \in X$, $r > 0$. *Then* $B_r(x_0)$ *is an open set.*

Theorem. *Let (X, d) be a metric space. We have the following*

1. *The sets $\varnothing$ and X are open.*
2. *If A_1, A_2 are open subsets of X, then $A_1 \cap A_2$ is open.*
3. *If $\{A_i\}_{i \in I}$ is a family of open subsets of A, then $\bigcup_{i \in I} A_i$ is open.*

Observation Condition 2. above is equivalent to

2' If $n \in \mathbb{N}$ and $A_i \subseteq X$ is **open** for all $1 \leq i \leq n$, then $\bigcap_{i=1}^{n} A_i$ is **open**.

Exercise 2 Show that an infinite intersection of open sets may not be open.